INCH-POUND

MIL-DTL-17/84B
w/AMENDMENT 2
13 April 2010

SUPERSEDING
MIL-C-17/84B
AMENDMENT 1
14 September 1987

DETAIL SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, FLEXIBLE, COAXIAL, 50 OHMS, M17/84-RG223

Inactive for new design after 13 August 1993. For new design use MIL-C-17/194.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17.

NOTE: This cable uses PVC material and is not to be used in enclosed environments or shipboard applications. The replacements that are to be used in enclosed areas or shipboard applications are referenced in the following table.

The Air Force has restricted use of PVC in aerospace and ground support applications.

Cables listed on the current QPL may continue to be manufactured and supplied for existing enclosed applications only for a period not to exceed 3 years from the date of this specification.

TABLE I. Cross-reference data.

Current Part or Identifying Number (PIN)	Replacement PIN
M17/84-RG223	M17/194-00001

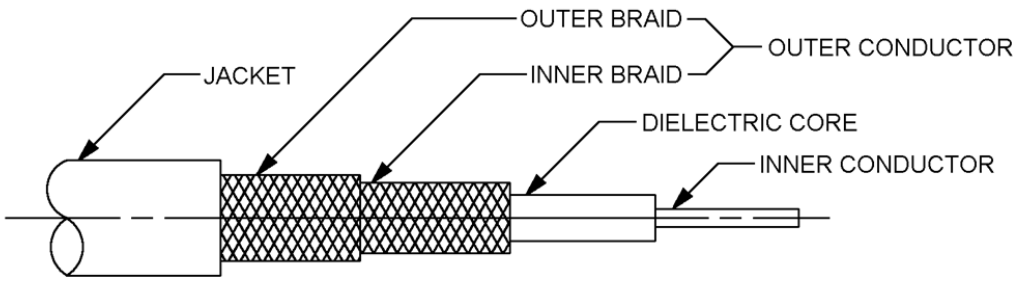


FIGURE 1. Configuration.

AMSC N/A

FSC 6145

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TABLE II. Description.

Components	Construction details
Inner conductor	Solid silver-coated copper wire. Overall diameter: 0.035 inch ± 0.001.
Dielectric core	Type A-1: Solid, polyethylene. Diameter: 0.116 inch ± 0.004.
Outer conductor	Double braid of AWG #36 silver-coated copper wire. Diameter: 0.176 inch maximum.
Inner braid	Coverage: 95.0% nominal Carriers: 16 Ends: 7 Picks/inch: 11.5 ± 10%
Outer braid	Coverage: 94.2% nominal Carriers: 16 Ends: 7 Picks/inch: 15.0 ± 10%
Jacket	Type IIa: PVC Diameter: 0.212 inch ± 0.004.

ENGINEERING INFORMATION

Continuous working voltage: 1,400 V rms, maximum.

Operating frequency: 12.4 GHz, maximum.

Velocity of propagation: 65.9 percent, nominal.

Power rating: See figure 2.

Operating temperature range: -40°C to +85°C.

Inner conductor properties:

DC resistance (maximum at 20°C): 0.897 ohm per 100 feet.

Elongation: 25 percent, minimum.

Tensile strength: Not applicable.

Engineering notes: This cable useful in general purpose, medium low temperature applications. (See connector series “TNC”, “BNC” and “SMA” in accordance with MIL-PRF-39012.)

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REQUIREMENTS

Dimensions, configuration, and description: See figure 1 and table II.

Environmental and mechanical:

Visual and mechanical examination:

Out-of-roundness: Not applicable.

Eccentricity: 10 percent, maximum.

Adhesion of conductors:

Inner conductor to core: 2 pounds, minimum; 20 pounds, maximum.

Aging stability: +98ºC $\pm$ 2ºC.

Stress crack resistance: Not applicable.

Outer conductor integrity: Not applicable.

Cold bend: -55ºC $\pm$ 2ºC.

Dimensional stability: +85°C $\pm$ 2°C.

Inner conductor from core: 0.062 inch, maximum.

Inner conductor from jacket: 0.125 inch, maximum.

Contamination: Applicable.

Bendability: Not applicable.

Flammability: Not applicable.

Weight: 0.041 pound per foot, maximum.

Electrical:

Continuity: Applicable.

Spark test: 5,000 V rms, +10%, -0%.

Voltage withstanding: 5,000 V rms, +10%, -0%.

Insulation resistance: Not applicable.

Corona extinction voltage: 1,900 V rms, minimum.

Characteristic impedance: 50 ohms $\pm$ 2.

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Attenuation: See figure 2.

Structural return loss: See figure 3.

Capacitance: 32.2 pF per foot, maximum.

Capacitance stability: Not applicable.

Capacitance unbalance: Not applicable.

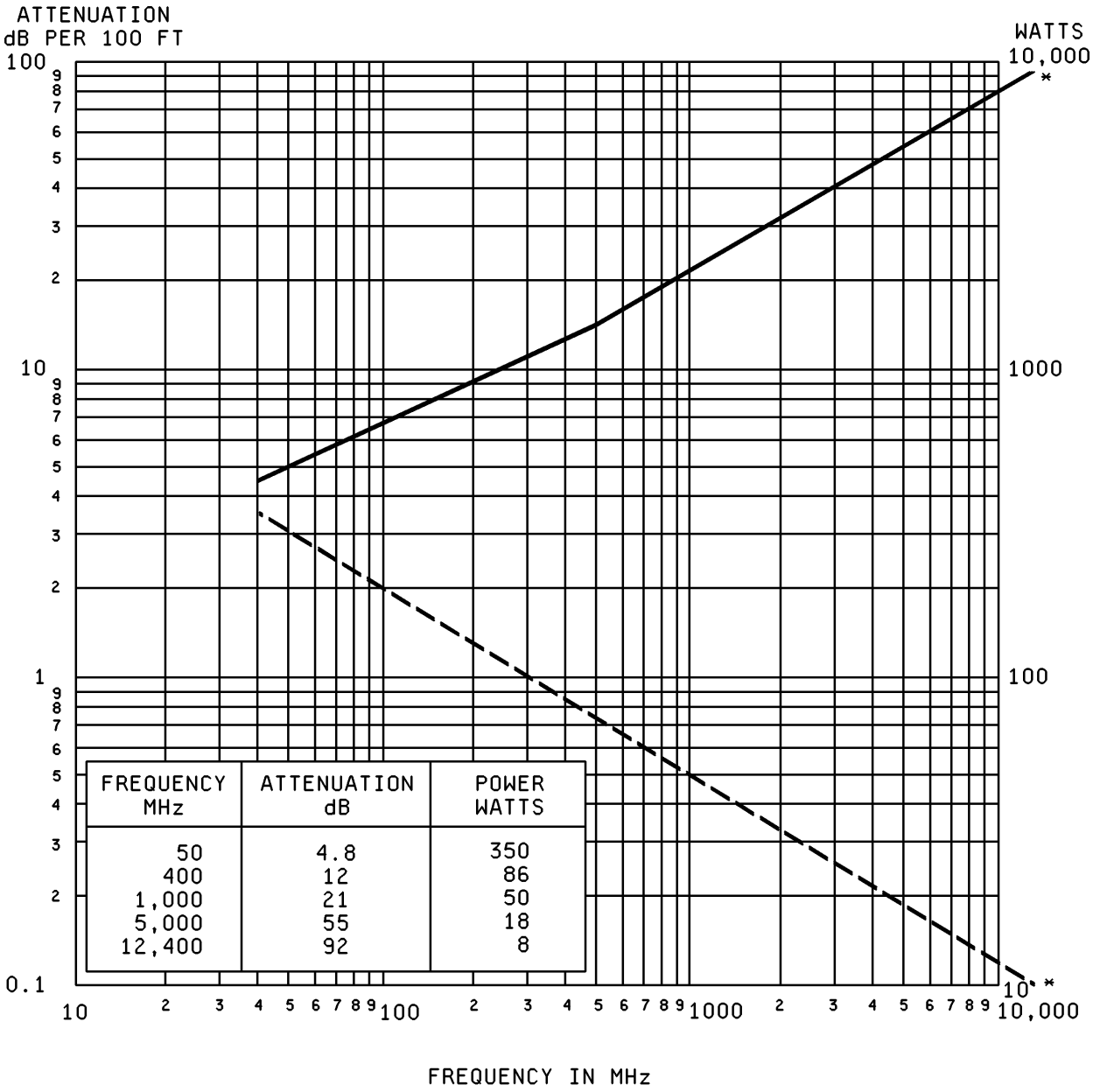
Transmission unbalance: Not applicable.

Mechanically induced noise voltage: Not applicable.

Time delay: Not applicable.

PIN: M17/84-RG223. See table I.

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FREQUENCY MHz	ATTENUATION dB	POWER WATTS
50	4.8	350
400	12	86
1,000	21	50
5,000	55	18
12,400	92	8

MAXIMUM ATTENUATION ———

MAXIMUM POWER ·— — — -

AT 25°C SEA LEVEL

* At 12,400 GHz.

FIGURE 2. Power rating and attenuation.

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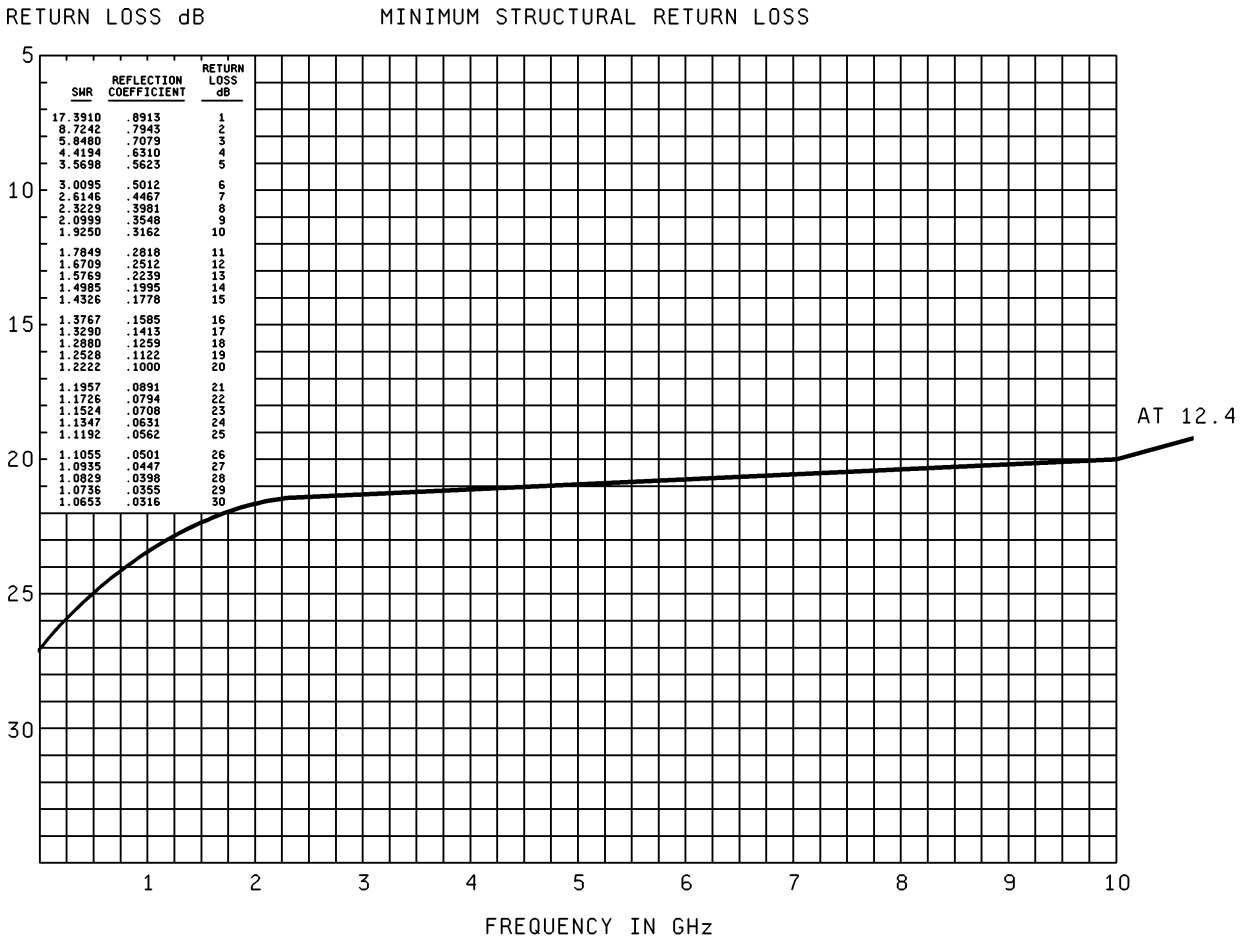


FIGURE 3. Structural return loss.

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Referenced documents. In addition to MIL-DTL-17, this document references the following:

MIL-PRF-39012
MIL-C-17/194

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 6145-2010-033)

Review activities:
Army – AR, AT, CR4, MI
Navy – AS, MC, OS, SH, TD
Air Force – 19, 99
DLA - IS

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